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Window regulator

Abstract

A window regulator installed on a door panel of a vehicle door. The window regulator includes a carrier plate that supports a window glass of the vehicle door, a guide rail that supports the carrier plate in an ascendable and descendible manner, a housing including a rail fitting portion to which an end portion of the guide rail is fitted, and a cover plate that closes a working hole of the door panel. The housing and the cover plate are integrally molded.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present patent application claims the priority of Japanese patent application No. 2023/104457 filed on Jun. 26, 2023, and the entire contents of Japanese patent application No. 2023/104457 are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a window regulator.

BACKGROUND ART

(3) A lower end drive-type window regulator is known which has a drive unit arranged at a lower end of a guide rail (see Patent Literature 1). The window regulator includes a carrier plate connected to a window glass of a vehicle, a guide rail to guide ascending and descending of the carrier plate, a wire to pull the carrier plate, a rotating drum that takes up and feeds out the wire by rotating, and a drum housing (casing) to which the lower end of the guide rail is attached and in which the rotating drum is housed. This type of window regulator is configured such that the drum housing with the guide rail attached thereto is attached to the door panel and the window regulator is thereby installed on the door panel.

CITATION LIST

Patent Literatures

(4) Patent Literature 1: JP 2013/124515A

SUMMARY OF INVENTION

(5) However, the known window regulator cannot be installed on the door panel if a working hole (service hole) of the door panel is located at the position of the drum housing. Therefore, it is necessary to install the window regulator so that the position of the drum housing does not overlap with the position of the working hole, which poses a problem in that the position of the window regulator with respect to the door panel is restricted.

(6) It is an object of the invention to provide a window regulator that can be installed with a simple configuration and without being restricted by the working hole.

(7) According to an aspect of the invention, provided is a window regulator installed on a door panel of a vehicle door, the window regulator comprising: a carrier plate that supports a window glass of the vehicle door; a guide rail that supports the carrier plate in an ascendable and descendable manner; a housing comprising a rail fitting portion to which an end portion of the guide rail is fitted; and a cover plate that closes a working hole of the door panel, wherein the housing and the cover plate are integrally molded.

Advantageous Effects of Invention

(8) The window regulator according to the aspect of the invention can be installed with the simple configuration and without being restricted by the working hole.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic front view showing a window regulator in an embodiment of the present invention and a vehicle door in which the window regulator is installed.
- (2) FIG. 2 is a front view showing the vehicle door.
- (3) FIG. 3A is a front view showing the window regulator when a window glass is fully closed.
- (4) FIG. 3B is a front view showing the window regulator when the window glass is fully open.
- (5) FIG. 4A is a cross-sectional view taken along line A-A' and showing a guide rail as shown in FIG. 3A.
- (6) FIG. 4B is a cross-sectional view taken along line A-A' and showing the area around a rail fitting portion as shown in FIG. 3A.
- (7) FIG. 5A is a perspective view showing a drum housing and a cover plate that are integrally molded.
- (8) FIG. 5B is a front view showing the drum housing and the cover plate that are integrally molded.
- (9) FIG. 5C is a plan view showing the drum housing and the cover plate that are integrally molded.

DESCRIPTION OF EMBODIMENTS

(10) A window regulator in an embodiment of the invention will be described below with reference to the appended drawings. The window regulator is a raising/lowering device that is installed in a vehicle door and raises/lowers a window glass on the vehicle door. In particular, the drum housing and the cover plate of this window regulator are molded integrally, allowing the window regulator to be installed without being restricted by a working hole. Hereinafter, an ascending/descending direction of the window glass will be simply referred to as the ascending/descending direction. In addition, left, right, front, rear, up and down used in the following description are as defined in each drawing. In the present embodiment, the ascending/descending direction coincides with the up-down direction of the window regulator, and a vehicle width direction coincides with a front-rear direction of the window regulator. In addition, a direction orthogonal to the ascending/descending direction and the vehicle width direction is a left-right direction of the window regulator.

(11) (Configuration of a Vehicle Door)

(12) Here, before describing a window regulator **21**, a vehicle door **1** in which the window regulator **21** is installed will be described. As shown in FIGS. 1 and 2, the vehicle door **1** is composed of an inner panel **11** as a door panel on the interior side of the vehicle and an outer panel (not shown) on the exterior side of the vehicle, and a door internal space is formed between the inner panel **11** and the outer panel.

(13) A window opening **16**, which is opened and closed by a window glass **W**, is formed on upper portions of the inner panel **11** and the outer panel. Then, three working holes **17a**, **17b** and **17c** to introduce various functional components into the door internal space are formed on a lower portion of the inner panel **11**. Hereinafter, the working hole **17a** to introducing the window regulator **21** into the door internal space will be referred to as the “first working hole”.

(14) (Configuration of the Window Regulator)

(15) As shown in FIGS. 1, 3A and 3B, the window regulator **21** is installed on the inner panel **11** of the vehicle door **1**, and includes a window regulator main body **31**, and cover plate **32** that supports the window regulator main body **31** and closes the first working hole **17a**.

(16) As shown in FIGS. 3A and 3B, the window regulator main body **31** includes a carrier plate **41** that supports the window glass **W**, a guide rail **42** that is provided along the ascending/descending direction and supports the carrier plate **41** in an ascendable and descendible manner, an ascending-

side wire **43** and a descending-side wire **44** that pull the carrier plate **41**, a drive unit **45** arranged at a lower end of the guide rail **42** to drive the ascending-side wire **43** and the descending-side wire **44**, and a pulley **46** fixed to an upper end of the guide rail **42** to change the direction of the ascending-side wire **43**. That is, the window regulator main body **31** is a wire driving-type window regulator to raise and lower the carrier plate **41** using the wires **43**, **44** and is also a lower end drive-type window regulator having the drive unit **45** arranged at the lower end of the guide rail **42**. The ascending-side wire **43** and the descending-side wire **44** are examples of the wire.

(17) The guide rail **42** is a long metal member extending along the ascending/descending direction and is curved so as to bulge in the vehicle width direction toward the outside of a vehicle cabin. As shown in FIG. **4A**, the guide rail **42** has a flat plate portion **51** extending in the ascending/descending direction, a right-side plate portion **52** and a left-side plate portion **53** that rise toward the front from left and right ends of the flat plate portion **51** that are ends in a lateral direction (a direction perpendicular to the ascending/descending direction), and a right flange portion **54** protruding to the right from a front end of the right-side plate portion **52**.

(18) As shown in FIGS. **3A** and **3B**, the ascending-side wire **43** is connected at one end to a rotating drum **62** (described later) of the drive unit **45**, is fed upward from the drive unit **45** to the pulley **46**, is redirected downward by the pulley **46**, and is then attached at the other end to the carrier plate **41**. On the other hand, the descending-side wire **44** is connected at one end to the rotating drum **62** of the drive unit **45**, is fed upward from the drive unit **45**, and is attached at the other end to the carrier plate **41**.

(19) The carrier plate **41** is a plate-shaped member formed of, e.g., a resin such as polyacetal, and is slidably supported by the right-side plate portion **52** of the guide rail **42**. The carrier plate **41** also has a fastening hole **41a** for attachment of the window glass **W**. The fastening hole **41a** is for bolt-fastening of a glass holder (not shown) which is fixed to the window glass **W**. The glass holder is attached to the fastening hole **41a** with a bolt, and the window glass **W** is thereby attached to the carrier plate **41** through the glass holder.

(20) The drive unit **45** has a drive motor **61**, the cylindrical rotating drum **62** that is rotationally driven by the drive motor **61** and takes up and feeds out (drives) the ascending-side wire **43** and the descending-side wire **44** by rotating, a reducer (not shown) to transfer rotation of the drive motor **61** to the rotating drum **62**, a drum housing **63** to which the lower end of the guide rail **42** is fitted and in which the rotating drum **62** is rotatably housed, and a motor housing **64** holding the drive motor **61**. The motor housing **64** incorporates the reducer and is fixed to the drum housing **63**. The motor housing **64** covers an opening of a drum housing portion **71** (described later) of the drum housing **63** in a state in which an output shaft of the reducer is connected to the rotating drum **62**. The drum housing **63** is an example of the housing.

(21) When the drive motor **61** is driven forward, the rotating drum **62** rotates forward and takes up the ascending-side wire **43** while feeding out the descending-side wire **44**. Thus, the carrier plate **41** is pulled by the ascending-side wire **43** and moves upward in the ascending/descending direction. As a result, the window glass **W** attached to the carrier plate **41** ascends. On the other hand, when the drive motor **61** is driven reverse, the rotating drum **62** rotates reverse and takes up the descending-side wire **44** while feeding out the ascending-side wire **43**. Thus, the carrier plate **41** is pulled by the descending-side wire **44** and moves downward in the ascending/descending direction. As a result, the window glass **W** attached to the carrier plate **41** descends. In this manner, the carrier plate **41** and the window glass **W** are raised and lowered along the guide rail **41**.

(22) As shown in FIGS. **5A** to **5C**, the drum housing **63** is formed of a resin such as polyacetal and is integrally molded with the cover plate **32**. The drum housing **63** has the drum housing portion **71** having a housing hole **71a** in which the rotating drum **62** is rotatably housed, a rail fitting portion **72** arranged above the drum housing portion **71** and fitted with the lower end of the guide rail **42**, and a first fixing portion **73**, a second fixing portion **74** and a third fixing portion **75** for fixing the motor housing **64** with screws. The first fixing portion **73** is formed on the rail fitting portion **72**,

and the second fixing portion **74** and the third fixing portion **75** are respectively arranged at the lower right and lower left of the drum housing portion **71** and are connected to the drum housing portion **71** through rib-shaped connecting portions **76**.

(23) The rail fitting portion **72** has a fitting hole **72a** which is open on the upper side and into which the lower end of the guide rail **42** is inserted from above. As shown in FIG. **4B**, the fitting hole **72a** has a shape complementary to the cross section of the guide rail **42**, and when the lower end of the guide rail **42** is inserted into the fitting hole **72a**, inner walls of the fitting hole **72a** facing the front and back surfaces (on the front and rear sides) of the flat plate portion **51** come into contact with the front and back surfaces of the flat plate portion **51** and the flat plate portion **51** is sandwiched and held in between, and also inner walls of the fitting hole **72a** facing the front and back surfaces (on the left and right sides) of the left-side plate portion **53** come into contact with the front and back surfaces of the left-side plate portion **53** and the left-side plate portion **53** is sandwiched and held in between. The lower end of the guide rail **42** is thereby fitted into the fitting hole **72a** and fixed. As a result, the lower end of the guide rail **42** is fixed to the drum housing **63**.

(24) As shown in FIGS. **5A** to **5C**, the cover plate **32** is formed of a resin such as polyacetal and integrally has a flat plate main body **81**, a rising portion **82** rising rearward from a circumferential end portion of the plate main body **81**, and a projecting portion **83** that projects outward (toward the side opposite to the plate main body **81**) from an end of the rising portion **82**. The plate main body **81** has the same shape as the first working hole **17a**, and a raised portion composed of the plate main body **81** and the rising portion **82** is fitted to the first working hole **17a**. Meanwhile, eight clips **84** for fastening to the inner panel **11** are attached to the projecting portion **83**. The cover plate **32** is attached to the inner panel **11** by fastening the projecting portion **83** to the inner panel **11** with the eight clips **84** while fitting the plate main body **81** and the rising portion **82** to the first working hole **17a**.

(25) Although not shown in the drawings, a sealant for sealing is applied between the projecting portion **83** and the inner panel **11**. This sealant improves the watertightness between the projecting portion **83** and the inner panel **11** and prevents water from entering the inside of the vehicle door through a gap between the cover plate **32** and the inner panel **11**. In this window regulator **21**, the cover plate **32**, which is integrally molded with the drum housing **63**, has this sealant.

(26) As described above, the cover plate **32** and the drum housing **63** are integrally molded. In particular, the cover plate **32** and the drum housing **63** are integrally formed by molding a resin such as polyacetal. The drum housing **63** is molded on the front of the plate main body **81**, and the drum housing **63** is thereby fixed to the plate main body **81**.

(27) (Operation of Installing the Window Regulator **21**)

(28) Next, an operation of installing the window regulator **21** on the vehicle door **1** will be described. In the operation of installing the window regulator **21**, first, the window regulator main body **31** is introduced into the door internal space through the first working hole **17a**. After that, the projecting portion **83** of the cover plate **32** is fastened to the inner panel **11** using the eight clips **84** while fitting the plate main body **81** and the rising portion **82** of the cover plate **32** to the first working hole **17a**. After fastening the projecting portion **83** to the inner panel **11**, the upper end of the window regulator main body **31** is fixed to the inner panel **11**. Then, the operation of installing the window regulator **21** ends.

(29) (Functions and Effects of the Embodiment)

(30) In the configuration of the embodiment described above, integrally molding the drum housing **63** and the cover plate **32** allows the window regulator **21** to be installed on the inner panel **11** with a simple configuration without being restricted by the working hole **17a**.

(31) In addition, integrally molding the drum housing **63** and the cover plate **32** also eliminates the need for use of special fastening members (bolts and nuts, etc.) to fix the drum housing **63** to the cover plate **32** and also eliminates the need work of attaching the drum housing **63** to the cover plate **32**, hence, the manufacturing cost of the window regulator **21** can be reduced.

(32) In addition, integrally molding the drum housing **63** and the cover plate **32** also allows the fastening point of the drum housing **63** to the inner panel **11** to be away from the rail fitting portion **72**. That is, since the drum housing **63** and the cover plate **32** are integrally molded, the drum housing **63** is fastened to the inner panel **11** though the cover plate **32**, hence, the fastening point of the cover plate **32** to the inner panel **11** serves as the fastening point of the drum housing **63** to the inner panel **11**. Thus, the fastening point of the drum housing **63** to the inner panel **11** can be away from the rail fitting portion **72**. Therefore, it is possible to reduce the stress applied to the fastening point when moment with respect to the guide rail **42** or the rail fitting portion **72** as a fulcrum occurs. As a result, damage or breakage of the drum housing **63** or the inner panel **11** can be suppressed when moment with respect to the guide rail **42** or the rail fitting portion **72** as a fulcrum occurs due to rattling of the window glass **W**, etc. Furthermore, since the number of fastening points can be increased as compared to when the drum housing **63** is directly fastened to the inner panel **11**, the stress applied can be dispersed to each fastening point and the load applied to each fastening point can be reduced.

(33) Furthermore, instead of integrally molding the guide rail **42** directly with the cover plate **32**, the drum housing **63**, to which the lower end of the guide rail **42** is fitted, is integrally molded with the cover plate **32**, hence, it is possible to replace the guide rail **42** with one having a different curvature or length. Thus, appropriately replacing the guide rail **42** with one having a different curvature or length allows to be compatible with the vehicle door **1** having a different curvature or size. That is, the same components except the guide rail **42** can be used for vehicle doors **1** having different curvatures or sizes, thereby reducing the manufacturing cost of the window regulator **21**.

(34) In addition, since the drum housing portion **71** and the rail fitting portion **72** are formed on the drum housing **63**, the positional accuracy regarding the positional relationship (the relative positions) of the drum housing portion **71** and the rail fitting portion **72** can be improved.

Other Embodiments

(35) Although the embodiment of the invention has been described, the invention according to claims is not limited to the embodiment described above. Further, please note that not all combinations of the features described in the embodiment are necessary to solve the problem of the invention. The invention can be appropriately modified and implemented without departing from the gist thereof.

(36) For example, in the embodiment described above, the invention is applied to the drum housing **63** with the drum housing portion **71** formed thereon. However, it is not limited thereto. That is, the invention may be applied to a housing that has the rail fitting portion **72** but does not have the drum housing portion **71**.

(37) In addition, in the embodiment described above, the drive motor **61** is arranged on the front side of the inner panel **11** (on the exterior side of the vehicle). However, the drive motor **61** may be arranged on the rear side of the inner panel **11** (on the interior side of the vehicle).

(38) In addition, in the embodiment described above, the invention is applied to the housing arranged at the lower end of the guide rail **42**. However, the invention may be applied to a housing arranged at the upper end of guide rail **42**. That is, the invention may be applied to a housing on which the rail fitting portion **72** fitted with the upper end of the guide rail **42** is formed.

(39) In addition, in the embodiment described above, the invention is applied to the single-rail window regulator **21** with a single guide rail **42**. However, the invention may be applied to a double-rail window regulator **21** with plural guide rails **42**, etc.

(40) In addition, in the embodiment described above, the invention is applied to the lower end drive-type window regulator **21**. However, the invention may be applied to other window regulators having the drive unit **45** at different positions, such as delta-type, banjo-type, or self-propelled type.

(41) In addition, in the embodiment described above, the window regulator **21** installed in the left front door (i.e., the left front side door) has been described as an example. However, it is not

limited thereto. That is, the invention may be applied to the window regulator **21** installed in the right side door, or the invention may be applied to the window regulator **21** installed in the rear side door.

REFERENCE SIGNS LIST

(42) **1** VEHICLE DOOR **11** INNER PANEL **17a** WORKING HOLE **21** WINDOW REGULATOR
32 COVER PLATE **41** CARRIER PLATE **42** GUIDE RAIL **43** ASCENDING-SIDE WIRE **44**
DESCENDING-SIDE WIRE **62** ROTATING DRUM **63** DRUM HOUSING **71** DRUM HOUSING
PORTION **72** RAIL FITTING PORTION W WINDOW GLASS

Claims

1. A window regulator installed on a door panel of a vehicle door, the window regulator comprising: a carrier plate that supports a window glass of the vehicle door; a guide rail that supports the carrier plate in an ascendable and descendible manner; a housing comprising a rail fitting portion to which an end portion of the guide rail is fitted; and a cover plate that is attached to the door panel and closes a working hole formed in the door panel, wherein the housing and the cover plate are integrally molded, and wherein the housing is attached to the door panel through the cover plate.
 2. The window regulator according to claim 1, further comprising: a wire to pull the carrier plate; and a rotating drum that drives the wire by rotating, wherein the housing further comprises a drum housing portion to house the rotating drum.
 3. The window regulator according to claim 1, wherein the housing and the cover plate are integrally formed by molding a resin.
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